Section 4.3

30.

a. That is, $\Phi(z) = 0.91$

By the table, $z \approx 1.34$

b. since $0.09 + 0.91 = 1$. the values of them are symmetric about $z=0$.

Thus $z = -1.34$ for 9th percentile.

c. $\Phi(z) = 0.75$. By the table, $z \approx 0.67$.

d. Since $0.25 + 0.75 = 1$. the values of them are symmetric about $z=0$.

Thus $z = -0.67$ for 25th percentile.

e. $\Phi(z) = 0.06$. By the table, $z \approx -1.56$.

44. a. $P(\mu - 1.5\sigma \le X \le \mu + 1.5\sigma)$

$$= P\left(\frac{\mu - 1.5\sigma - \mu}{\sigma} \le Z \le \frac{\mu + 1.5\sigma - \mu}{\sigma}\right)$$

$$= P(-1.5 \le Z \le 1.5)$$

$$= \Phi(1.5) - \Phi(-1.5)$$

$$= 0.9332 - 0.0668$$

$= 0.8664$

b. $P(X < \mu - 2.5\sigma \text{ or } X > \mu + 2.5\sigma)$

$= 1 - P(\mu - 2.5\sigma \le X \le \mu + 2.5\sigma)$

$= 1 - P\left(\frac{\mu - 2.5\sigma - \mu}{\sigma} \le Z \le \frac{\mu + 2.5\sigma - \mu}{\sigma}\right)$

$= 1 - P(-2.5 \le Z \le 2.5)$

$= 1 - (\Phi(2.5) - \Phi(-2.5))$

$= 1 - (0.9938 - 0.0062)$

$= 0.0124$

c. $P(\mu - 2\sigma \le X \le \mu - \sigma \text{ or } \mu + \sigma \le X \le \mu + 2\sigma)$

$= P(\mu - 2\sigma \le X \le \mu - \sigma) + P(\mu + \sigma \le X \le \mu + 2\sigma)$

$= P(-2 \le Z < -1) + P(1 \le Z \le 2)$

$= \Phi(-1) - \Phi(-2) + \Phi(2) - \Phi(1)$

$= 0.1587 - 0.0228 + 0.9772 - 0.8413$

$= 0.2718$

48.

a. Since $\Phi(-1.72)+\Phi(1.72)=1$ due to the symmetry of standard normal distribution. $\Phi(1.72)=0.9573$

Thus $\Phi(-1.72)=1-\Phi(1.72)=0.0427$

Similarly. $\Phi(-0.55)=1-\Phi(0.55)=0.2912$

Thus $P(-1.72\le Z\le -0.55)=\Phi(-0.55)-\Phi(-1.72)$

$=0.2485$

b. $P(-1.72\le Z\le 0.55)=\Phi(0.55)-\Phi(-1.72)$.

from (a). we obtain $\Phi(-1.72)=0.0427$.

Thus $P(-1.72\le Z\le 0.55)=0.7088-0.0427$

$=0.6661$

So it is not necessary according to the symmetry about $z=0$ of standard normal curve.

53. a. For all $p=0.5$, $p=0.6$ and $p=0.8$, $np>10$ and $n(1-p)\geq 10$. Thus we can use normal approximation.

① $p=0.5$, $\mu=np=12.5$, $\sigma=\sqrt{np(1-p)}=2.5$.

$$P(15\leq X\leq 20)=P\left(\frac{15-0.5-12.5}{2.5}\leq Z\leq\frac{20+0.5-12.5}{2.5}\right)$$

$$=P(0.8\leq Z\leq 3.2)$$

$$=\Phi(3.2)-\Phi(0.8)$$

$$=0.2112.$$

② $p=0.6$, $\mu=np=15$, $\sigma=\sqrt{np(1-p)}=2.45$

$$P(15\leq X\leq 20)=P\left(\frac{15-0.5-15}{2.45}\leq Z\leq\frac{20+0.5-15}{2.45}\right)$$

$$=P(-0.20\leq Z\leq 2.24)$$

$$=\Phi(2.24)-\Phi(-0.20)$$

$$=0.5668$$

③ $p=0.8$, $\mu=np=20$, $\sigma=\sqrt{np(1-p)}=2$.

$$P(15\leq X\leq 20)=P\left(\frac{15-0.5-20}{2}\leq Z\leq\frac{20+0.5-20}{2}\right)$$

$= P(-2.75 \le Z \le 0.25)$

$= \Phi(0.25) - \Phi(-2.75)$

$= 0.5957$.

Exact probabilities: $P(15 \le X \le 20) = B(20; 25, p) - B(14; 25, p)$.

For $p = 0.5, 0.6, 0.8$, They are: 0.212, 0.577, 0.573.

b. ① $P(X \le 15) = P(Z \le \frac{15+0.5-12.5}{2.5}) = P(Z \le 1.2)$

$= \Phi(1.2)$

$= 0.8849$.

② $P(X \le 15) = P(Z \le \frac{15+0.5-15}{2.45}) = P(Z \le 0.20)$

$= \Phi(0.20)$

$= 0.5793$.

③ $P(X \le 15) = P(Z \le \frac{15+0.5-20}{2}) = P(Z \le -2.25)$

$= \Phi(-2.25)$

$= 0.0122$

Exactly probabilities: $P(X \le 15) = B(15; 25, p)$.

For $p = 0.5, 0.6, 0.8$, They are: 0.885, 0.575, 0.017

c. ① $P(X \geq 20) = P\left(Z \geq \frac{20-0.5-12.5}{2.5}\right) = P(Z \geq 2.8)$

$= 1 - \Phi(2.8)$

$= 0.0026$

② $P(X \geq 20) = P\left(Z \geq \frac{20-0.5-15}{2.45}\right) = P(Z \geq 1.84)$

$= 1 - \Phi(1.84)$

$= 0.0329$

③ $P(X \geq 20) = P\left(Z \geq \frac{20-0.5-20}{2}\right) = P(Z \geq -0.25)$

$= 1 - \Phi(-0.25)$

$= 0.5987$.

Exactly probability: $P(X \geq 20) = 1 - B(19; 25, p)$

For $p = 0.5, 0.6, 0.8$, They are: 0.002, 0.029, 0.617.

56. For a general normal percentile with $N(\mu, \sigma)$.

Since $Z = \frac{X-\mu}{\sigma}$. Thus for corresspondi̇ng standard

$(100p)$th percentile, $P(Z \leq z_{1-p}) = P(X \leq z_{1-p}\sigma + \mu)$

Thus the relationship is $P(Z \leq z_{1-p}) = P(X \leq z_{1-p}\sigma + \mu)$

Section 4.4

59. a. $E(X) = \frac{1}{\lambda} = 1$.

b. $\sigma = \sqrt{\frac{1}{\lambda^2}} = 1$.

c. $P(X \le 4) = F(4;1) = 1 - e^{-4} = 0.982$.

d. $P(2 \le X \le 5) = (1-e^{-5}) - (1-e^{-2}) = 0.129$

67. $\mu = 24 = \alpha\beta$. $\sigma = 12 = \sqrt{\alpha\beta^2}$. Thus $\alpha = 4$, $\beta = 6$

a. $P(12 \le X \le 24) = F(4;4) - F(2;4)$

$= 0.424$.

b. $P(X \le 24) = F(4;4) = 0.567 > 0.5$

Thus the median is less than 24.

c. $F(\frac{c}{6}; 4) = 0.99$ By the table, $\frac{c}{6} = 10$.

Thus $c = 60$. That is, the 99th percentile is 60.

d. $P(X > t) = 0.005$. Thus $P(X \le t) = 1 - P(X > t)$

$= 0.995$.

That is. $F(\frac{t}{6}, 4) = 0.995$. By the table, $\frac{t}{6} = 11$. Thus $t = 66$.

70. That is, $F(x;\lambda) = 1 - e^{-\lambda x} = p$.

Thus $x = -\frac{\ln(1-p)}{\lambda}$

For the median, $p = 0.5$. $x = \frac{\ln 2}{\lambda}$.